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Influence of oxygen and carbon on performance of superpermeable membranes

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Abstract

In order to understand the applicability of superpermeable membranes for tritium recovery from fusion exhaust gas containing impurities, the influence of oxygen and carbon on the performance of Nb and V membranes was examined by laboratory scale experiments. No significant degradation due to oxygen was observed if the membrane temperature was sufficiently high, i.e. above 700 K, and the bulk oxygen concentration was below 1.5 at.%. It was also possible to avoid the influence of carbon deposition by the operation at high temperatures through the conversion of carbon deposited layer into carbide layer.
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1. Introduction

Metallic membranes covered by chemisorbed non-metallic impurities such as oxygen and sulfur can be *superpermeable* for suprathermal hydrogen particles (e.g. hydrogen atoms and ions) [1]. Suprathermal hydrogen particles can penetrate into the bulk through such adlayers of non-metallic impurities, while the re-emission of hydrogen molecules is strongly impeded. Therefore, almost all hydrogen particles sticking to the surface permeate through the membrane, irrespective of its thickness and temperature. It is possible to separate the permeating hydrogen particles from any impurity gases including He and compress them at the downstream side [2]. Panels of such membranes could be installed along the walls of pumping ducts with atomizes or discharge system to recover a major part of D/T before cryogenic pumps to decrease the tritium load on the pumps (see Fig. 5 in reference [3]).

In exhaust gases from fusion reactors, however, not only He but also other impurities such as carbon and oxygen are contained in the forms of hydrocarbons,

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CO, H_2O, and so on. Therefore, the formation of thin films of metal oxides and deposited carbon is possible. In contrast to the chemisorbed impurities in *monolayer*, such *films* could impede the penetration of suprathermal hydrogen particles [4–6] and radically reduce permeation rate.

The purpose of the present study is to understand the appropriate operation condition of superpermeable membranes to avoid the formation of surface films of metal oxides and deposited carbon. Niobium and vanadium were selected as candidates of membrane materials because of (i) high diffusivity of hydrogen isotopes in the bulk, (ii) small recombination rate constant of hydrogen even at clean surface, and (iii) high chemical reactivity resulting in thermally and chemically stable chemisorbed impurity layers [7]. The decomposition of V and Nb oxides at elevated temperatures by dissolution of oxygen into the bulk was investigated by means of X-ray photoelectron spectroscopy (XPS) and Auger electron spectroscopy (AES). The influence of oxygen accumulation in the bulk on the performance of the membranes was investigated by atom- and gas-driven permeation experiments. Absorption experiments were also carried out for carburized Nb and V specimens to examine the influence of carbide formation. The durability of the membranes under the conditions of fusion exhaust gas was discussed.

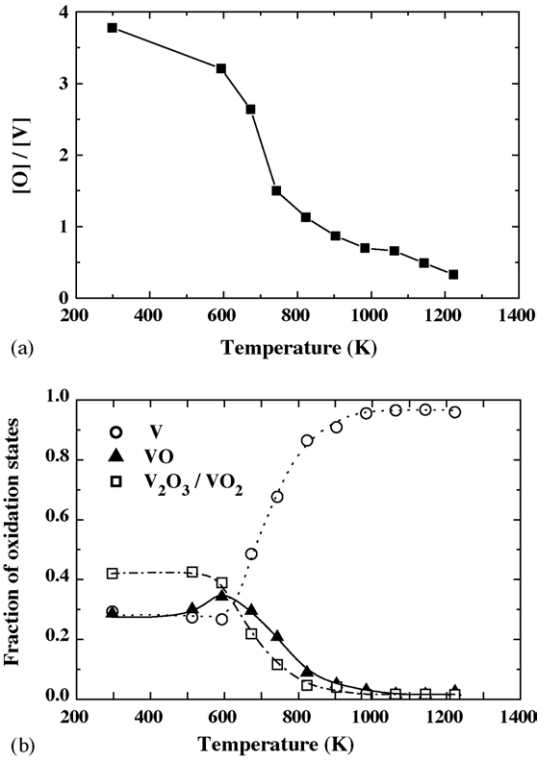


Fig. 1. Change in atomic ratio of oxygen to V ([O]/[V]) (a) and oxidation state of V (b) by heating V specimen in vacuum.

2. Influence of oxygen

Solubility of oxygen in group 5 metals is high in comparison with other metals [8]. For example, the solubility in V is 3 at.% at 473 K and increases with temperature. Hence, oxide films could be decomposed by the dissolution of oxygen in the bulk at temperatures where diffusion of oxygen is thermally enhanced.

In the case of V, oxygen in an oxide film started to diffuse into the bulk around 700 K. Fig. 1 shows the result of surface analysis of V specimen after heating in vacuum. In this case, a sheet type specimen of polycrystalline V was mechanically polished and then introduced in an ultra-high vacuum (UHV) chamber. The pressure of residual gases was lower than 2×10^{-7} Pa. The specimen was heated at a given temperature for 600 s, and the decomposition of so-called natural oxide film was examined by XPS. The change in oxidation state of V was analyzed by deconvolution of V 2p photoelectron spectra, and the atomic ratio of oxygen to V ([O]/[V]) was evaluated from the intensity ratio between O 1s and V 2p photoelectron peaks [9]. It is apparent that [O]/[V] started to decrease at around 700 K. The conversion of V oxides into the metallic state was also initiated at 700 K and almost completed at 900 K. It should be emphasized that [O]/[V] did not reach zero even in the temperature range above 900 K. Schiechl and Winkler [6] examined the oxidation of V(1 0 0) surface by XPS under 6.6×10^{-6} Pa of O_2 at various temperatures and reported that V_2O_3 was formed below 873 K, whereas (5×1)-O superstructure was observed at higher temperatures. In addition, the absorption rate of atomic hydrogen significantly increased in the course of this change in the surface state [6]. By taking account of their observations, the oxygen remaining on the surface of the present V specimen above 900 K is considered to be in the chemisorption state. Similar dissolution of oxygen resulting in the formation of chemisorbed oxygen layer was observed

also for Nb as described below. Such chemisorbed oxygen layer could be responsible for superpermeation. The operation temperature should be higher under higher oxygen partial pressure to enhance the diffusion of oxygen to the bulk.

Oxygen dissolved in the bulk is gradually accumulated during the operation, and its concentration finally exceeds the solubility limit. Therefore, the influence of oxygen accumulation on the performance of Nb and V membranes was examined with an UHV apparatus described elsewhere [10]. A membrane of polycrystalline Nb or V (0.1 mm in thickness) was doped with oxygen step by step up to 1.5 at.% by heating the membrane in O_2 gas. First, a steady-state O_2 pressure (in the range of 10^{-6} to 10^{-2} Pa) was established at room temperature, and then the membrane was heated to 1273 K. The pressure of O_2 dropped owing to the absorption by the membrane, and the amount of oxygen absorbed was evaluated from the extent of pressure drop and the duration of heating. After introduction of oxygen, the membrane was heated in vacuum ($<1 \times 10^{-7}$ Pa) at 1500 K to ensure the uniform distribution of oxygen over its thickness. The atom-and gas-driven permeation experiments were carried out in a temperature range from 473 to 1413 K at upstream H_2 pressures of 0.1–10 Pa. Hydrogen atoms were produced by incandescent filaments made of Ta. The sticking coefficients of hydrogen atoms and molecules, α_H and α_{H_2}, were evaluated from the permeation rate in a steady state. It was difficult to measure the incident flux of atoms onto the membrane surface, and hence the incident flux was roughly evaluated by taking account of various parameters such as the surface area of filaments and dissociation probability of hydrogen molecules on the filament surfaces. Hence, the absolute value of α_H obtained in the present study contains uncertain constant coefficient. Nevertheless, the observed tendency of the dependence of α_H on bulk oxygen concentration was reliable, because the reproducibility of permeation experiments was quite good if the filament temperature was kept constant.

The results for Nb membrane are shown in Fig. 2 as an example. The sticking coefficient of hydrogen atoms, α_H, was completely independent of the bulk oxygen concentration, C_O, while α_{H_2} decreased with increasing C_O. It should be emphasized that C_O exceeded the solubility limit in a low temperature region; the solubility limit of oxygen in Nb is less than

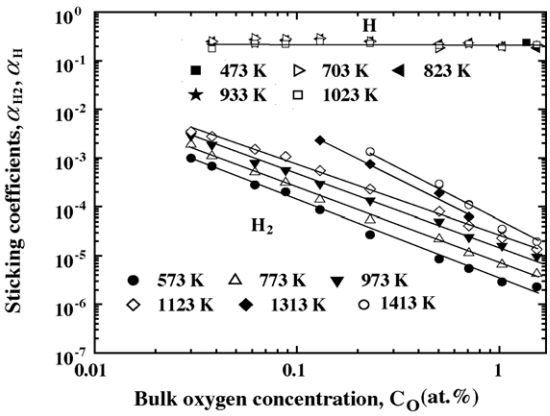


Fig. 2. Influence of bulk oxygen concentration on sticking coefficients of hydrogen atoms, α_H, and molecules, α_{H_2}, on Nb surface.

1.5 at.% below 973 K [8]. The sticking coefficient of hydrogen atoms is strongly reduced by the formation of oxide films as mentioned above [4,6]. It is therefore plausible that the formation of oxide films can be avoided even at a bulk oxygen concentration exceeding the solubility limit if heat treatment is appropriate. In order to confirm this expectation, the surface of Nb sheet containing 2 at.% of oxygen prepared in the above-mentioned manner was analyzed by AES in a separate UHV chamber at temperatures from 723 to 1573 K; the measurement was started at 1573 K, and then the specimen temperature was reduced step by step. The pressure of residual gases was lower than 5×10^{-7} Pa. Before heating, the specimen surface was covered by so-called natural oxide film as shown in Fig. 3 because of exposure to air before introduction into the chamber. This oxide film was decomposed during the measurement at 1573 K. An example of Auger spectrum obtained at 723 K is also shown in Fig. 3. This spectrum agreed well with that for a Nb surface covered by chemisorbed oxygen given by Sürgers et al. [11] as expected. These observations indicate that accumulation of oxygen in the bulk does not degrade the superpermeability. Similar results were obtained for V. The correlation between α_{H_2} and C_O shown in Fig. 2 would be discussed in a separate paper.

The membranes became brittle by oxygen absorption up to 1.5 at.%. Nevertheless, they were not broken during the permeation experiments, in which the maximum pressure difference between upstream and downstream sides was 10 Pa.

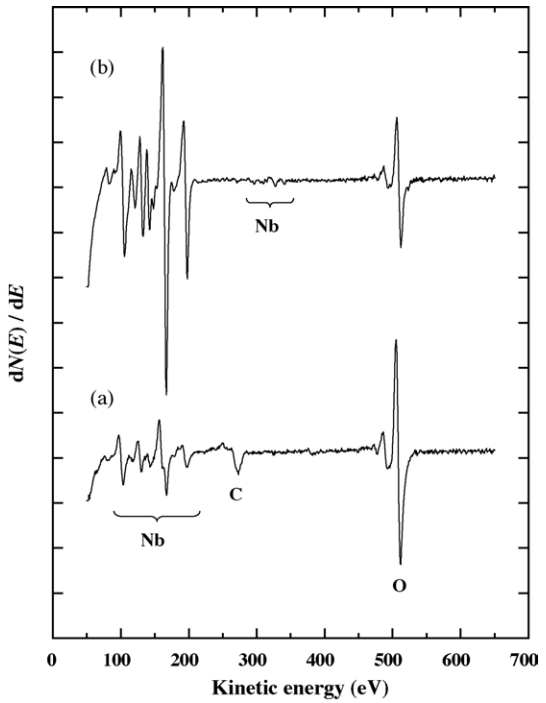


Fig. 3. Auger spectrum of Nb specimen containing 2 at.% of oxygen obtained before heating (a) and at 723 K (b).

3. Influence of carbon

As mentioned in Section 1, the deposition of carbon strongly reduces the permeation rate [5]. Nevertheless, such negative influence of carbon film could be eliminated also by high temperature heating. In contrast to oxygen, the solubility of carbon in group 5 metals is rather small [12]. Hence, high temperature heating leads to the formation of carbides such as M_2C and MC, where M = V, Nb and Ta. The permeability of hydrogen isotopes through carbide layers has not been evaluated quantitatively. Nevertheless, the results of experiments described below showed that the permeability in the carbide layers is rather high.

A panel of Nb sheet was installed in the large helical device (LHD) of National Institute of Fusion Science (NIFS) during the 5th cycle of LHD operation to examine the sorption capability of fuel particles in the edge plasma. Then, detailed absorption tests for atomic hydrogen were carried out in a separate UHV chamber, in which the panel was heated at 1123 K for 1.2 ks prior to the tests [13]. The analyses by means of AES and X-ray diffraction method showed that the panel surface was covered by the carbide layer consisting of NbC and Nb_2C (several hundreds nm in thickness) owing to the carbon deposition during the exposure to LHD plasma and the reaction during the subsequent heat treatment. Nevertheless, the panel showed the pumping rate of $10^{20}\,H\,m^{-2}\,s^{-1}$ for atomic hydrogen at temperatures above 473 K, and this pumping rate was comparable to the value obtained before the installation in LHD [13].

Preliminary tests to examine the hydrogen permeability through V carbide layer have been also carried out. In this case, sheet type V specimens (10 mm × 10 mm × 0.5 mm) were polished mechanically and then carburized in a mixture gas of H_2 and CH_4 at 973 K to form V_2C layer whose thickness was several hundreds nm. After the preparation of thin Pd layer (100 nm) by vacuum evaporation to enhance the surface reaction, the absorption of H_2 was examined at 773 K. The absorption rates of carburized specimens were comparable to those of uncarburized specimens. This observation indicates that the penetration of hydrogen atoms formed on the surface of Pd layer was not impeded by the diffusion process through V_2C layer. It is therefore appropriate to consider that V_2C layer does not retard the permeation of suprathermal hydrogen particles.

4. Durability against impurities in fusion exhaust gas

According to Poschenrieder et al. [14,15] who analyzed the exhaust gas of ASDEX by means of mass spectroscopy, the main impurity species were hydrocarbons such as methane, ethane and ethylene, CO and water vapor under wide variety of wall conditions, although the partial pressures of these impurity gases were sensitively dependent on the materials of plasma-facing components and the method of wall conditioning. Since these impurity gases are considered to be important also in future fusion reactors, the attention was focused on these species in this section.

The influence of water vapor appears to be most serious among the above-mentioned impurity species. Although the interaction between water vapor and the surfaces of group 5 metals covered by oxygen has been scarcely examined, the sticking coefficient of O_2 has been measured in a wide temperature range for Nb;

the value of sticking coefficient was ca. 0.01 at 745 K and 0.2 at 1428 K [16]. By assuming that the sticking coefficient of water vapor is comparable to that of O_2, the value around 900 K, for example, is given to be ca. 0.1. If the bulk oxygen concentration is limited up to 1.5 at.% to avoid the damage by the embrittlement, the lifetime of the membrane whose thickness is 0.1 mm is evaluated to be 320 ks (90 h) under the partial pressure of water vapor of 10^{-4} Pa. The lifetime would be shorter under higher partial pressure of water vapor. Therefore, wall conditioning such as Ti evaporation and boronization [14,15] is required to minimize the partial pressure of water vapor.

It is also necessary to develop techniques to remove oxygen from the membrane in situ in the pumping ducts. Oxygen in Nb can be removed as Nb oxides by heating above 2100 K [16] and as CO above 1673 K [17]. If the membrane is combined with a discharge system, removal of oxygen is possible at more moderate temperatures, e.g. 1113 K, by hydrogen ion sputtering through a combination of continuous surface segregation and sputtering [18]. No reliable technique has been developed to remove oxygen from V.

The presence of alkanes such as methane and ethane is not a critical issue, if the superpermeable membrane is used with incandescent filaments. This is because the reaction coefficients (probability of decomposition at a single collision) of these hydrocarbons are sufficiently small. By extrapolating the date obtained by other researchers at temperatures above 1400 K [19,20], the reaction coefficients of methane and ethane on clean Nb surface at 900 K were evaluated to be 2×10^{-6} and of 3×10^{-5}, respectively. It is also known for Ta that the presence of chemisorbed oxygen layer reduces the reaction coefficients by from 4 to 6 orders of magnitude [21]. Alkenes such as ethylene are much more reactive than alkanes. Nevertheless, the reaction coefficients of alkenes are also reduced by the presence of chemisorbed oxygen. For example, the reaction coefficient of ethylene which was almost unity on a clean Ta surface was reduced to ca. 10^{-4} at 1673 K by chemisorbed oxygen [21]. In addition, a major portion of deposited carbon can be transformed into metal carbides being permeable for hydrogen isotopes at elevated temperatures as described above. If the average reaction coefficient of hydrocarbons is 10^{-6} and the total pressure of hydrocarbons is 10^{-3} Pa, it takes more than 20 years to convert 1% of Nb into carbides when the thickness of the membrane is 0.1 mm. Much shorter lifetime is expected if suprathermal hydrogen particles are generated by discharge because hydrocarbon radicals are also produced. Therefore, the atomizers are preferable to use under high partial pressures of hydrocarbons.

The reaction of CO on the surfaces of group 5 metals covered by non-metallic impurities has not been investigated in detail. Nevertheless, the rate of reaction appears to be rather small, because a Nb membrane could be operated without degradation at 800–950 K for 720 ks (200 h) with Ta filaments in the presence of CO of 6×10^{-4} Pa [3]. Further study is, however, necessary for the quantitative evaluation of influence of CO.

5. Conclusions

Oxide films formed on V and Nb surfaces were decomposed by heating in vacuum above 700 K owing to diffusion of oxygen into the bulk, and layers of chemisorbed oxygen remained. The accumulation of oxygen in the bulk led to no noticeable degradation in superpermeability up to 1.5 at.%. Carbides of Nb and V were permeable for hydrogen. These results indicate that the operation of Nb and V superpermeable membranes is possible in fusion exhaust gas containing oxygen and carbon if the membrane temperature is sufficiently high to enhance the oxygen diffusion and carburization reactions.

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